

SAKON NAKHON, Thailand

WMO: 483560

Lat: 17.160N

Lon: 104.133E

Elev: 172

StdP: 99.28

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	14.5	7.7	6.7	17.8	9.3	7.4	18.3	5.3	24.6	4.7	24.3	0.3	120	0.295

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.4	37.4	25.0	36.1	25.1	34.8	25.3	27.6	32.2	27.2	31.8	26.9	31.5	1.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.5	29.4	26.1	22.0	29.1	25.8	21.5	28.9	88.9	32.3	87.3	32.0	85.9	31.6	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.5	4.0	3.4	10.1	39.2	1.7	1.2	8.8	40.1	7.8	40.8	6.8	41.5	5.6	42.3		
			8.8	28.7	1.6	1.0	7.7	29.4	6.7	30.0	5.8	30.6	4.7	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	22.9	25.0	27.8	29.6	29.2	28.9	28.2	27.9	27.8	27.2	25.4	22.9
	DBStd	3.16	2.71	3.28	3.22	2.60	1.89	1.49	1.54	1.30	1.23	1.47	2.20	2.76
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	6	2	1	0	0	0	0	0	0	0	0	0	2
	CDD10.0	6173	401	419	553	588	596	567	566	556	534	532	462	400
	CDD18.3	3137	144	187	294	338	337	317	307	298	284	273	212	144
	CDH23.3	33231	1191	1802	3368	4197	3952	3661	3307	3086	2902	2696	1894	1176
	CDH26.7	13552	396	793	1689	2210	1781	1487	1169	997	964	997	702	367
Wind	WSAvg	0.8	1.0	1.1	1.1	0.8	0.7	0.7	0.8	0.7	0.5	0.7	0.8	0.9
Precipitation	PrecAvg	1615	6	18	48	80	247	299	219	348	263	74	9	4
	PrecMax	2013	47	94	140	127	455	449	364	999	501	268	59	45
	PrecMin	1364	0	0	0	15	136	148	56	107	108	2	0	0
	PrecStd	210	13	27	41	34	87	95	102	209	117	69	16	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	36.4	38.7	39.6	37.6	35.6	35.1	34.0	33.6	33.3	34.2	33.0
		MCWB	22.2	22.6	23.2	25.1	26.3	26.6	26.7	26.9	26.4	25.2	24.5	23.7
	2%	DB	31.7	34.6	37.0	37.9	35.8	34.3	33.4	32.9	32.6	32.5	32.6	31.4
		MCWB	22.0	22.8	24.0	25.4	26.5	26.6	26.4	26.5	26.0	24.9	24.0	22.7
	5%	DB	30.4	33.1	35.5	36.3	34.5	33.3	32.4	31.9	31.8	31.9	31.6	30.2
		MCWB	21.6	22.9	24.1	25.6	26.3	26.4	26.3	26.3	25.8	24.7	23.6	22.0
	10%	DB	29.0	31.4	33.9	34.8	33.1	32.3	31.3	30.9	31.0	31.1	30.4	28.8
		MCWB	20.9	22.6	24.2	25.5	26.1	26.3	26.2	26.2	25.6	24.3	23.0	21.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	25.6	26.5	27.4	27.8	28.0	27.7	27.7	27.6	26.7	25.9	24.8
		MCDB	30.2	31.8	33.5	34.7	33.5	32.4	32.0	31.6	30.7	31.0	30.9	30.8
	2%	WB	23.5	24.7	25.7	26.8	27.3	27.4	27.1	27.2	27.0	26.1	25.2	23.6
		MCDB	28.9	31.3	32.9	34.2	32.9	31.9	31.2	31.0	30.6	30.2	30.1	29.5
	5%	WB	22.5	24.0	25.2	26.4	26.9	27.0	26.7	26.7	26.6	25.6	24.7	22.8
		MCDB	28.1	30.2	32.4	33.6	32.3	31.5	30.6	30.3	30.0	29.6	29.3	28.6
	10%	WB	21.7	23.2	24.7	25.9	26.6	26.7	26.4	26.4	26.2	25.2	24.0	22.0
		MCDB	27.8	29.3	31.7	32.9	31.6	30.8	30.1	29.8	29.4	29.1	28.4	27.7
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.8	10.5	9.4	7.5	6.4	5.7	5.7	6.1	7.8	9.9	11.2
		MCDBR	12.7	13.1	11.8	10.8	9.1	7.6	7.1	7.1	7.3	8.4	10.4	11.6
		MCWBR	4.4	4.2	3.5	3.0	2.7	2.4	2.4	2.6	2.3	2.5	3.3	4.0
	5% WB	MCDBR	10.9	11.0	10.3	9.6	7.9	6.9	6.4	6.6	6.5	7.4	8.7	10.7
	MCWBR	4.4	4.1	3.7	3.3	2.9	2.6	2.6	2.7	2.5	2.5	3.1	4.0	
Clear-Sky Solar Irradiance	taub	0.457	0.569	0.662	0.628	0.511	0.470	0.457	0.468	0.506	0.519	0.456	0.454	
	taud	2.146	1.855	1.683	1.799	2.146	2.289	2.321	2.278	2.159	2.087	2.237	2.178	
	Ebn at Noon	824	747	694	721	800	826	838	836	805	781	820	813	
	Edh at Noon	146	205	250	224	156	134	130	137	153	160	132	138	
All-Sky Solar Radiation	RadAvg	4.85	5.14	5.13	5.51	5.28	4.88	4.53	4.51	4.64	4.99	4.89	4.65	
	RadStd	0.29	0.33	0.22	0.27	0.35	0.37	0.43	0.33	0.36	0.35	0.18	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page